UNIT 1 - Introduction to ML

Evolution of Machine Learning

Machine Learning (ML) is the process of learning a model that can be used in Prediction based on data.

Prediction involves assigning a data item to one of the classes or associating the data item with a number.

→ In early days AI (Artificial Intelligence) opined that mathematic logic was the ideal vehicle for building AI systems.

→ DL (Deep Learning) is an branch of ML

→ Perceptron was the earliest popular ML tool and forms various DL architectures including MLP (Multilayer perceptron); Convolutional neural networks (CNNs) and recurrent neural networks (RNN's).

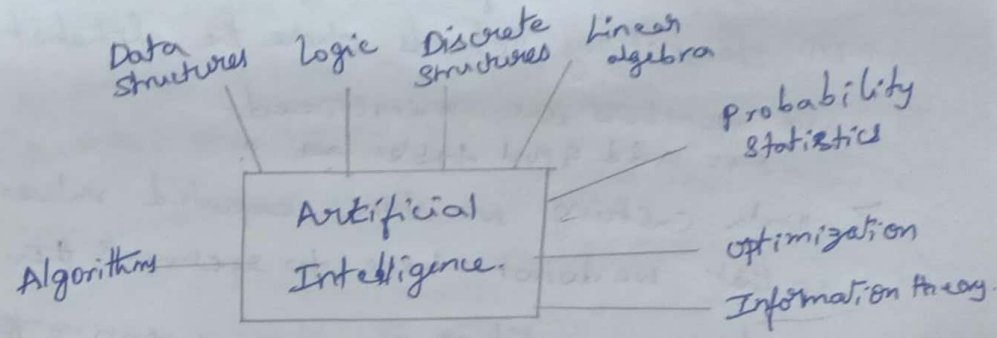
→ In early 21st Century and the current view is the AI is deep learning and deep learning is AI.

→ Graphics processing units (GPUs) platforms like Tensor flow and Pytorch ~~success~~ stories of CNN.

ML along with DL become of state-of-the art subject. ANN backbone of DL.

The importance of other background topics as:

* In ML, vector and vector spaces are in Linear algebra.
* The role of probability and statistics Explained in statistical ML.
* optimization is essential in training neural networks
* Information theoretic concepts like entropy and divergence are essential to DNN



Paradigms For ML

There are different ways or paradigms for ML.

1. Learning by rote
2. Learning by abduction
3. Learning by induction and reinforcement Learning.
4. Learning by deduction

Learning by Rote:

→ It is a form of learning that is popular in elementary Schools where the alphabet and numbers are memorized.

Ex: add & Mul tables ~~are~~.

→ Data Caching we store computed values so that we donot have to recompute them later.

→ caching computation is more expensive than recall.

Ex: Chess masters lot of time memorizing great times, of the past.

→ Rote learning teaches how to think in chess.

Learning by Deduction:

This type of learning is based on reasoning that is truth preserving.

Given A, if A then B (A → B), deduce B.

we can use B along with if B then C (B → C) to deduce C. A and A → B are true, then B is true.

1. It is raining
2. If it rains, the roads get wet.
3. If a road is wet, it is slippery.

From (1) and (2) we can infer using deduction that (4) the roads are wet.

this deduction then used with (3) to deduce & learn that (5) the roads are slippery.

If stmt (1) (2) (3) are true, then stmt (4) (5) are also true

Learning by Abduction: Making a probable conclusion from what you know.

We infer A from B and (A → B). This is not truth preserving like in deduction as both B and (A → B) can be true and A can be False.

Ex:

1. An aeroplane is a flying object (aero → flying obj
2. A is a flying object.

(1) & (2) we infer using abduction that A is an aeroplane. This kind of reasoning may lead to incorrect conclusions. for Ex 'A' could be a bird or a kite.

Learning by Induction

this is most popular and effective form of ML.

Learning from examples:

1. classification: consider hand written digits

Each row has 15 Examples of each of the digits.

the problem is to learn an ML model using such data to classify a new data pattern. This is example of supervised Learning.

Regression: contrary to classification, there are several prediction applications where the labels come from possibly infinite set.

Ex. The share value of a stock could be a +ve real number. The stock may have different values at a particular time and each of these values is a real no. This is typical regression.

The practical need here is to predict the share value of a stock at a future time instance based on past data in the form of examples.

[To predict continous target variable based on input features.

Learning from observations:

observations are also instances like examples but they are different because observations need not be labelled. We cluster or group the observations into a small number of groups. Such grouping is performed with help of clustering alg. that assign similar patterns to the same group.

Ex.

0 0 0
1 1 1
3 3 3

Reinforcement Learning

Def: RL is subset of ML focused on training agents to make decisions in complex, uncertain environments to maximize cumulative rewards.

Key components: [How to make decisions to acheive

i) Agent: The decision-maker that takes actions in the environment.

Ex: Computer prog playing the game

ii) environment: the External world that responds to agent actions.

3. Actions: Decisions made by agent.
Ex: moving left or right

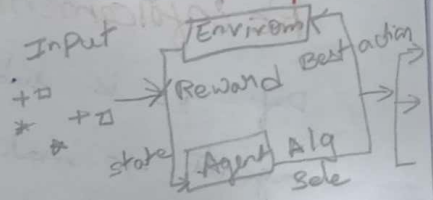
4. Rewards: Feed back signals indicating the desirability of the agent's actions.
Ex: like points

5. State: The current situation or status of the environment.
Ex: current game situation
Possition of paddle.

Characterstics:

1. Trial and error learning: Agents learn through Experimentation.

2. Feed back loop: Agents receive rewards or penalities for actions.



Types of Data:

In general data can be categorized into

1. Categorical
2. Numerical.

Categorical:

→ This type of data can be nominal or ordinal

→ Nominal data can be no order among the elements.

Ex: Coloring hair domain {brown, red, black}

→ Ordinal data can be order among the elements

Ex: Employee number D{1, 2, ... 100}.

Numerical:

→ The domain of values of the data type could be a set/subset of integers or set/subset of real numbers.

Feature No	Attribute	Type of Data	Domain
1	Diagnosis	Nominal	{Malignant, Benign}
2	ID Number	Ordinal	{8670, 917897}
3	Perimeter_Mean	Numerical	{43.79, 188.5}
4	Area_Mean	Numerical	{143.5, 2501}
5	Smoothness_Mean	Numerical	{0.05263, 0.1634}

Table 1.1 Different types of data from cancer DB

classification context, class label data can have the domain $\{-1, +1\}$

$-1 \rightarrow$ label of negative class

$+1 \rightarrow$ label of positive class

Each pattern is represented as vector of feature values.

Ex: A data item corresponding to a patient with ID 92751 is rep by 5-D vector

(Benign, 92751, 47.92, 181.0, 0.05263)

Benign is value of feature 1, Diagnosis.

Illy 3rd entry 47.92 to feature 3 and

So on.

Diagnosis is nominal

ID number is ordinal attribute

Diagnosis is dependent feature. Remaining Four feature are independent features.

ED → distance measures, compare image.
Cosine → similarity, Ex: Reason exp.

Matching

1. Matching is an important activity in ML.
2. It is both used in both supervised learning & in learning
3. Matching carried out by using proximity measure which can be distance measure
4. Two data items u and v dimension vectors.
5. A popular distance measure is Euclidean distance and a popular similarity measure the straight line distance between 2 points in MD space. Cosine of the angle between vectors.

$$d(u,v) = \sqrt{\sum_{i=1}^{l} (u(i) - v(i))^2}$$

Cosine similarity is given by measure cosine of angle between 2 vectors in a MD space

$$\cos(u,v) = \frac{u^t v}{\|u\| \|v\|}$$

$u^t v$ is the dot product between u & v.

$\|u\| \rightarrow$ euclidean distance between u and origin. It is called euclidean norm.

Important application of matching ML are in.

1. Finding the Nearest Neighbor of a pattern:

Let x be an d-dimensional pattern vector..

Let $\mathcal{X} = \{x_1, x_2 \ldots x_n\}$ collection of n data vectors. The nearest neighbor of x from $\mathcal{X}$, denoted by $NN_x(\mathcal{X})$ is x_j if

$$d(x, x_j) \leq d(x, x_i) \ \forall \ x_i \in \mathcal{X}$$

where there is a tie $x_p \in \mathcal{X}$ and $x_q \in \mathcal{X}$ are the nearest neighbors of x, we can break the tie arbitrarily or choose either of two to be the nearest neighbor of x.

2. Assigning to a set with the Nearest Representative:

Let $c_1, c_2 \ldots c_k$ be k sets with $x^1, x^2, \ldots x^k$ as their respective representatives. A pattern x assigned to c_i if $d(x, x^i) \leq d(x, x^j)$ for $j \in \{1, 2 \ldots k\}$

This idea useful in clustering.

Stages in Machine Learning:

Application Domain → Data acquistion

↓

Feature Engineering [Preproc + Repres]

↓

Model selection → choose a Model ← Domain knowledge

↓

Model learning → Train the model ← Training data

↓

Model Evaluation → Validate the model ← evaluation data

↓

Model Prediction → learn the model ← Test data

↓

Model Explanation → Explain the model ← Expert feedback

Data acquistion:

Machine learn from data that give them.

It is important to collect valuable data so that Model can find correct patterns.

Feature engineering:

→ Putting together all the data have & randomizing it.

For that data evenly distributed, and ordering.
→ cleaning the data to remove unwanted data, missing values, rows, colours, duplicate values, data type conversion etc.

Choose a Model:
It is imp to choose a model which is relevant to the task at hand.

Train the Model:
In training pass the prepared data to ML Model to find patterns and make predictions.

Validate the model:
After training model, have to check to see how it's performing. This is done by testing the performance of the model on previously unseen data.

Learn the model: once created and evaluated model, if its accuracy can be improved in any way. This is done by tuning the parameters present in Model.

Explain Model: In end model on unseen data to make predictions accurately.

Feature Engineering:

It is combination of preprocessing + Data Rep.

Data Preprocessing:
Raw data needs to be updated used by ML.
Common problem:

1. Missing data: It occurs inability to measure a feature value or due to unavailability.
Different schemes used for missing values.

* Nearest Neighbor: Let x be l-dimensional data vector that has its ith component $x(i)$ missing. Let $\mathcal{X} = \{x^1, x^2, \ldots x^n\}$ n traing vectors $x^j \in \mathcal{X}$ be nearest neighbor of x based on remaining $l-1$ (excluding ith) components.
Predict the value $x(i)$ to be $x^j(i)$.

* larger neigborhood: use KNN of x to predict missing value x_i.

$$\frac{1}{K} \sum_{j=1}^{K} x_j(i)$$

cluster data and locate the nearest cluster.

cluster the Training data and locating the cluster to which x belongs on the remaing $l-1$ components.

Let x its ith value missing belong to cluster c^q. Let μ^q be centroid of c^q. Then predict value of $x(i)$ is μ_i^q, the ith component of μ^q.

Ex

$$\begin{bmatrix} 5.1 & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 4.7 & 3.2 & 1.3 & 0.2 \\ 4.6 & 3.1 & 1.5 & 0.2 \\ 5.0 & 3.6 & 1.4 & 0.2 \end{bmatrix}$$

Let use imagine that two values are missing to obtain

$$\begin{bmatrix} NA & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 4.7 & 3.2 & 1.3 & 0.2 \\ 4.6 & 3.1 & 1.5 & 0.2 \\ 5.0 & 3.6 & 1.4 & 0.2 \end{bmatrix}$$

We assume that this data is from same cluster. we can compute the mean in each case and it use it to predict the missing value. The mean of the available 4 value in the first column is 4.8.

$$\begin{bmatrix} 4.8 & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 4.7 & 3.2 & 1.3 & 0.2 \\ 4.6 & 3.325 & 1.5 & 0.2 \\ 5.0 & 3.6 & 1.4 & 0.2 \end{bmatrix}$$

Mean Squared Error:

$$\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$$

In the first case, instead 5.1, our estimed is 4.8.

In the second case, for the value 3.1 our estimate is 3.325

$$\frac{(5.1-4.8)^2 + (3.1-3.325)}{2} = \frac{0.09 + 0.00050625}{2}$$

$$= 0.04525$$

2. Data from different Domains:

consider for example, classification of objects into one of two classes: adult or child. Let the objects be represented by height in meteres and weight in grams. Consider an adult represent by the vector (1.6, 75000) and a child represented by the vector (0.6, 5000), where heights of the adult and child in metres are 1.6 & 0.6 and weights of adult and domain of weight is [2000, 200000]. Domain of height [0.5, 2.5]

Euclidean distance between adult & child

$$\sqrt{(1.6-0.6)^2+(75000-5000)^2}$$

$$=\sqrt{1+4.9\times10^9} \approx 70000$$

Cosine of angle between adult & child vector

$$\frac{0.96+375\times10^6}{\sqrt{25000000}\cdot 36\times\sqrt{5625000002.56}} \approx 1.0$$

There are two popular normalization schemes:

* Scaling using the range:
On any categorical feature the values of 2 patterns either match or mismatch and contribution to the distance either zero (0) match or 1 (mismatch).

$$\frac{|x^i(P)-x^j(P)|}{\text{Range } P}$$

* Standardization:
The data is normalized so that it will have 0 (zero) mean and unit variance.

3. Outliers in the Data:
An outliers is a data item that is either noisy or erroneous.

A data item is usually called an outlier if it
→ Assumes values that are far away from those of are data items
→ Deviates from normally behaving item.

Outliers occur due to different reasons:
1. Noisy measurements: It is malfunction & lead recording noisy data
2. Erroneous data entry: Data can occur at the data entry level itself.
Ex: very common for spelling mistakes to occur when names are entered.
3. Evolving systems: It is possible to encounter data items in sparse regions during evolution system.
4. Very naturally: Instead of viewing an outlier as a noisy or unwanted data item, it may be better to view it as something useful.

Data Representation:

Representation is an important step in building ML models.

Representation of Data items:

The active and state-of-the-art paradigm for ML is statistical machine learning.

Each data item is represented as a vector.

We consider addition of vectors in computing the mean or centroid of a collection of vectors, multiplication of a vector by scalar.

Some of the difficulties associated with ML using such high-dimensional data vectors are:

* Computation time increases with the dimensionality
* Storage space requirement increase with the dimensionality.

Feature selection & Feature Extraction Example.

Cars = make, model, year, mileage, color, attributes horse power, engine size, fuel efficiency.

F.S = most imp features = year, mileage, horsp, Eng size

F.E = Performance score = HP + ES.

age = curr year − year

Feature selection: Process of selecting a subset of most relevant features from larger set.

Let $F = \{f_1, f_2, \ldots f_L\}$ be set of L features, like to select a subset F_l of F having $l (< L)$ features s.t F_l maximizes.

Feature Extraction: Process of transforming raw data into a set of features that represent data in meaningful.

Linear Schemes.

$$h_j = \sum_{i=1}^{L} \alpha_{ij} f_i$$

Principal components: The data set of n vectors in an L-dimensional space.

Non-negative matrix factorization (NMF)

When the data is non-negative, it is possible that PCs have negative entries.

Model Selection:

Selection of model to be used to train ML

Ex: some features are numerical & others are categorical, then classifiers based on perceptrons and support vector machine are not suitable.

Model learning:

This step depends on size & type of training data.

→ A subset of labelled data is used as training data.

→ Another subset used for model validation.

Ex: Decision tree suited for transparency.

If height of object is larger than 5 feet, it is likely to be an adult and not a child.

Model Evaluation:

Also called as model validation. This step requires earmarked data called validation data.

ML model works on training data. Then we say that model is well.

Model Prediction:

Testing the model that is learnt and validated.

It is used for prediction because both classification and regression tasks are predictive tasks.

Ex: It is like a doctor who predicts and makes a diagnosis when a new patient come in.

Model Explanation:

This step explain to an expert or a manager why a decision was taken by ML model.

This step helps explicit or implicit feedback from the user to further improve the model.

Explanation is important in DL.

Search and Learning:

Search had a specific role in conventional AI
It used in problem solving, theorem proving, planning and knowledge-based systems.

- Exact search is popular in DB for answering queries, in OS for operation like grep and symbol table.
- Search is important in learning a classification model and regression
- Inference is search in logic & probability.

Ex: Information theory we search for entropy.

Data Sets:

Data sets for classification

1. MNIST Hand Written Data set (Modified National Institute of Standards and Technology)

   i) There are 10 classes and each digit is viewed as image size 28x28 (784) pixels. each pixel range 0 to 255.

   ii) There are around 6000 digits as training patterns and around 1000 test patterns in each class and class label is also provided for each of digits

2. Fashion MNIST data set

   i) It is data set of Zalando's article images consisting of training set 60,000 ex and test set 10,000 examples.

   ii) 28x28 grey scale image.

   iii) It has same image size and structure of training and testing splits as MNIST

3. Olivetti Face Data set:

   i) consists of 10 different images each of 40 distinct subjects. For some subjects the images were taken at different times, w. varying the lighting, facial expressions.

   ii) All the images were taken against a dark homogeneous background with the subject in an upright, frontal position

   iii) Each image is of size 64x64 = 4096

   iv)

Data sets for Regression:

1. Boston Housing Data set:

→ It has 506 patterns

→ Each patterns is a 13-dimensional vector

2. Airline Passengers Data set:

→ this data set provides monthly total of US airline passengers from 1949 to 1960

→ this data set is taken from an inbuilt data set of Kaggle.